Project Framing

Hospital Chatbot for Color-Coded Clinical Pathways

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Purpose

This pack states what the project is doing: Part 1 (NLP acuity) is built; Part 2 is protocol-mapped colour pathways; Bayesian fusion is out of scope.

1 Project map

One-sentence summary

A hospital chatbot that maps a patient's chief complaint to a SATS acuity level and colour, then attaches a concrete clinical pathway (destination, time target, actions)—without requiring a TEWS+Bayesian fusion stack at intake.

Architecture (implemented)

```
Chief complaint
  -> Medical gate
  -> OpenMed NER (optional)
  -> BioBERT (5-class acuity)
  -> Softmax -> acuity c-hat in {1..5}
  -> C = f_SATS(c-hat)   % 1 Red, 2 Orange, 3 Yellow, 4/5 Green
  -> Pathway protocol P(C)
```

Part 1 — done

Component	Location
Fine-tuned BioBERT (baseline + SMOTE)	`fine-tuned-biobert/`
ML API (predict, gate, OpenMed, ...)	`curatio/server/ml/`
Express gateway	`curatio/server/`
React demo + presentation UI	`curatio/client/`
Holdout eval (∼99.92% colour accuracy)	`results/eval_outputs/`
Results slides	`results/slides/`

Colour mapping in code: `curatio/server/ml/predict.py` (`SATS_BY_ACUITY`).

Part 2 — designed

Colour today is a label; pathways add destination, $T_{\max}$, and actions. TEWS calculator and Bayesian fusion are deferred / out of scope.

Explicit non-goals

- No Bayesian network implementation
- No three-layer fusion controller (NLP + TEWS + Bayes)
- No claim that BioBERT "already does Bayesian fusion" — it does text→acuity classification; pathways attach deterministically to colour

2 Research gaps

1. **Acuity discontinuity** — colour does not drive care after the desk; answer with pathway object $P(C)$.
2. **LMIC / Ghanaian chief-complaint NLP** — local text→SATS colour via BioBERT + Curatio.
3. **Green Tag Burden** — Green pathway diversion to OPD / Minors / Polyclinic.
4. **Constrained classifier vs open LLM** — fixed 5-class head + deterministic maps.
5. **Text-only pre-triage** — NLP acuity before vitals; TEWS nurse-side later.
6. **Minority Red safety** — baseline vs SMOTE; report L1 recall.
7. **Colour must mean a pathway** — not only a badge.

3 Ten research questions

RQ1 Can unstructured chief complaints map to SATS acuity 1–5 with usable accuracy?

RQ2 Does correct acuity imply correct colour under fixed f_{SATS}?

RQ3 How should colours become pathways (destination, $T_{\max}$, actions)?

RQ4 Can a chatbot deliver pre-triage acuity and pathway guidance?

RQ5 How well does the system reject non-clinical input?

RQ6 Does OpenMed NER improve interpretability without replacing acuity?

RQ7 How does imbalance affect Level-1 (Red) recall?

RQ8 How overconfident is softmax, and what about nurse oversight?

RQ9 Can Green diversion be specified to reduce ED congestion?

RQ10 What LMIC architecture uses NLP acuity without TEWS+Bayes at intake?

Formal Part 1: Input $X \to \hat{c} \in \{1, \ldots, 5\} \to C = f_{\text{SATS}}(\hat{c})$.
Formal Part 2: $C \mapsto P(C) = (T_{\max}, \text{dest}, \text{actions}, \text{escalation})$.

4 Pathways design (no Bayesian)

Colour	$T_{\max}$	Destination	Actions (summary)
Red	0 min	Resuscitation	Code Red; bypass registration
Orange	10 min	Acute bed	Countdown; escalate if unclaimed
Yellow	60 min	Urgent ED stream	Standing orders in parallel
Green	Routine	OPD / Minors / Polyclinic	Divert from acute ED
Blue	Dignity	Mortuary pathway	Silent social-work protocol

5 Mathematics scope

In scope: softmax acuity; f_{SATS}; pathway $P(C)$; evaluation metrics; medical gate.

Out of scope: TEWS sum engine; Bayesian $P(C_k \mid E)$; fusion max-urgency controller.

6 Thesis outline

- Ch.1 Introduction — problem, 10 RQs, scope (no Bayes)
- Ch.2 Literature — SATS, Green Tag, triage NLP
- Ch.3 Methodology — BioBERT, colour map, pathways
- Ch.4 Results — holdout metrics, calibration, scenarios
- Ch.5 Conclusion — RQ answers, future TEWS overlay (optional)

Location: `thesis/first-draft/`.

7 Roadmap

- Done: Part 1 software/eval; framing; findings; thesis first draft; no-Bayes decision
- Write: deepen Ch.2 from reading notes; expand bib
- Optional build: pathway card fields in API/UI
- Do not schedule: Bayesian networks / fusion controller